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ORIGINAL RESEARCH

Pose-driven human activity anomaly detection in a CCTV-like environment

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Abstract

Human activity anomaly detection plays a crucial role in the next generation of surveillance and assisted living systems. Most anomaly detection algorithms are generative models and learn features from raw images. This work shows that popular state-of-the-art autoencoder-based anomaly detection systems are not capable of effectively detecting human-posture and object-positions related anomalies. Therefore, a human pose-driven and object-detector-based deep learning architecture is proposed, which simultaneously leverages human poses and raw RGB data to perform human activity anomaly detection. It is demonstrated that pose-driven learning overcomes the raw RGB based counterpart limitations in different human activities classification. Extensive validation is provided by using popular datasets. Then, it is demonstrated that with the aid of object detection, the human activities classification can be effectively used in human activity anomaly detection. Moreover, novel challenging datasets, that is, BMbD, M-BMbD and JBMOPbD, are proposed for single and multi-target human posture anomaly detection and joint human posture and object position anomaly detection evaluations.

1 | INTRODUCTION

Human Activity Anomaly Detection (HAAD) is a challenging research topic in computer version. HAAD provides a binary label that informs whether human activity is normal or abnormal. HAAD yields many applications for modern society, such as surveillance systems, assisted living, human-to-machine interaction, robotics, and sports analysis [1, 2]. The first challenge is that different datasets may have different definitions for anomalies. The same action in the different environment sometimes have different normality labels, such as *walking-on-the-sidewalk* is normal, but *walking-on-grass* is abnormal. Another challenge is that human activities are complex temporal sequences of mixed gestures performed with various styles, attitudes, viewpoints, and light conditions. Moreover, human bodies can assume many different positions preserving their overall appearance. Last but not least, detecting the beginning and end of the activities is not trivial and requires ad-hoc strategies [3, 4].

This work focuses on video-based human CCTV-like monitoring and surveillance. In particular, we focus on considering human actions that posture-level semantic labels can describe, for example, running, bending, jumping, hands-waving, and sitting. Similarly, we focus on those normal/abnormal movements that belong to the posture-level semantic information. For example, a person falling, running or fighting in a scene where these events do not normally happen is considered abnormal. Similarly, critical objects positioned or unexpectedly handled are considered as abnormal events. Therefore, this work presents a novel system that explicitly addresses different abnormal activities related to the posture-level and target object position semantic. [1] It is based on pose-driven dee

p learning networks that rely on human pose detections and RGB data within target bounding boxes to perform Joint RGB-Pose based features extraction, classification and anomaly detection.

Unlike the traditional generative methods for video anomaly detection [5–9], we propose a multimodal algorithm for detecting abnormal behaviours and abnormal objects position. Our system consists of three parallel processing branches. The first extracts body pose information using a pose estimation algorithm and convert it into high-level features that are further processed by a neural network. The second processes raw

[1] Video demos at https://youtu.be/7_mcWCB76Ps

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RGB data within target bounding boxes to compensate for first branch failures' and enrich motion description. The third branch runs an object detector to collect objects labels and positions. The three branches extract features that can feed conventional machine learning models for anomaly detection. Since branch one and two are pre-trained on human action recognition tasks, the proposed system can simultaneously detect body movements and object positions anomalies while providing human action labels as an additional description. This architecture focuses on single targets data. However, it is simple to generalise multiple human targets and objects by using a tracking system.

Extensive simulations are provided to highlight the efficacy and robustness of the proposed system. Firstly, pose, and RGB data is used to train the body movement branches. After picking up the best model, we fuse the body movement branches with the object position branches for joint pose-RGB HAAD. Our proposed work relies on detected body landmarks, so the accuracy of extracted pose information is critical. Therefore, popular video datasets were considered for evaluation and selected depending on the quality of detected landmarks. Furthermore, performance evaluation is carried out on our three novel datasets, that is, the Body Movements based Dataset (BMbD), the Multi-target Body Movements based Dataset (M-BMbD) and the Joint Body Movements and Object Position based Dataset (JBMOPbD), specifically recorded within Intelligent Sensing Lab, Newcastle University. In particular, BMbD proposes single-target posture-related anomalies; M-BMbD considers multi- target posture-related based anomalies; JBMOPbD further considers single-target posture-related anomalies alongside multi-object position based anomalies. Comparative results over the state-of-the-art are also provided to highlight the efficacy of the proposed method.

In principle, relying on RGB data, we can detect anomalies coming from any source, such as people, cars, explosions and traffic. Moreover, RGB based methods can work in the wild and low-resolution situations. However, this work provides evidence that autoencoders based on RGB data are suboptimal in the case of posture-level HAAD with limited training data. When the scene semantic is given and known, that is, human posture-level actions in a CCTV-like scene, RGB-based methods do not explicitly consider this additional contextual knowledge. Therefore, as already mentioned, this work proposes considering human pose-driven networks, as human poses are assumed to contain most of the valuable information for posture-based HAAD. We demonstrate that pose data can boost RGB-based networks for action classification and anomaly detection. On the other hand, RGB data can instead compensate poses data in action recognition tasks. In short, merely relying on human poses and object bounding boxes, as in this work, the recognition system is explicitly injected with refined, non-redundant knowledge, which leads to better performance.

This work focuses on semi-supervised HAAD [10]. In other words, no abnormal instances are presented to the algorithm during the training phase to build up abnormal expectations. Therefore, the anomaly detection algorithm is trained on the most challenging case where only normal data is available. In summary, the contributions of this work are:

- The joint RGB data, skeleton and localisation information model for Human Activity Anomaly Detection (HAAD), which provides real-time action explanation of the human activity;
- Three novel datasets for challenging posture-related and object position HAAD in CCTV-like environments: BMbD, M-BMbD and JBMOPbD.

Section 2 summarises related works. Section 3 presents the novel RGB-Pose based networks with the aid of prior object information for HAAD. In Section 4, Simulations and results are provided. Moreover, conclusions are summarised in Section 5.

2 | RELATED WORK

2.1 | Human-related pose estimation

Motivated by skeleton-based human action recognition, [11, 12] and human tracking [13], our work relies on popular pose detectors, as it is found to be advantageous for analyzing the no-grid data (body joints). In particular, the pose detector OpenPose [14] is exploited as a pre-processing step to extract 25 human keypoints from each target in all frames. Human poses represent the main human action learning sources for the proposed method. The results can decrease the influence of external factors related to videos' quality, such as background illumination, camera's viewpoints, and pedestrians' appearances. All pedestrians are represented by pose graphs where the nodes mean the corresponding joints and the edges mean the relations between two nodes. The pose detectors provide frame-by-frame data for a variety of human targets. However, humans need to be tracked [15] from one frame to another to obtain consistent target's identities. This step makes sure that data from two or more similar targets, such as multiple humans, are correctly grouped over time based on the tracked identity. Thus, this step is called pose estimation instead of pose detection. Human targets have to be identified in each frame since the deep learning algorithm has to consider the human behaviours in the temporal domain.

2.2 | RGB-based and pose-based anomaly detection

State-of-the-art HAAD methods [16] mainly rely on RGB raw data to perform the task. In very recent work, Sultani et al. [6] presented a fully supervised RGB-based anomaly detection system. The authors defined multiple learning instances to estimate scores for each video frame, that is, measuring the abnormality rate. Thus, only frame-level anomalies are considered in these methods, neglecting any other semantic level, that is, pixel-level and target-level. In semi-supervised learning, an important role is played by autoencoder based methods.

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Hasan et al. [8] proposed a combination of CNN-based autoencoders to model the temporal evolution of HOG-HOF features. Autoencoder-based approaches exploit normal data for training. Therefore, testing clips that resemble training data will be reconstructed with low error, while deviations from the training data are expected to be reconstructed with higher error. Thus, it is possible to identify abnormalities by analyzing the reconstruction error. Hasan et al. also provide pixel-level anomaly detection. However, it still does not provide any insights into what the abnormal pixel would represent, whether a human target performing unexpected actions or a car crashing, explosions or other abnormal events [8]. Chong et al. [9] proposed a similar approach but opting for combinations of CNN and LSTM for both Spatio-temporal encoding and decoding. However, this work does not provide insights into the nature of the detected anomalies. The other recent approaches are provided [17, 18], where background subtraction methods are used as a pre-processing step of raw data. It ensures that the system can focus only on moving objects. However, background subtraction methods fail on cluttered background scenes and in the case of pan-tilt-zoom camera recording. As opposite, human pose detection is robust to these challenges.

In [19], a pose-based HAAD is proposed, which is based on clustering-based soft-assignment and Dirichlet Process-based Mixture models. Similarly, in [20], a pose-based HAAD algorithm is presented, which leverages the MPED-RNN (single encoder dual decoder) architecture. However, both the algorithms in [19, 20] do not perform human body motion analysis. In contrast, deep-learning-based features are extracted from RGB and pose data in the proposed work, leading to simultaneous HAAD. Other remarkable works [21–26] share similar common limitations, such as the anomaly detection is performed at the pixel level, not considering which semantic region contains the pixel.

In this work, the above-mentioned autoencoder-based methods in [8], and [9] have been implemented as state-of-the-art baselines for semi-supervised HAAD to allow performance comparison on the proposed datasets. In section 4, we demonstrate that these methods fail in detecting the anomalies contained in the proposed datasets.

2.3 | Object detection

Object detection is always the core task in computer vision. An effective object detection algorithm can detect the key objects information from the whole frame in real-time. In our tasks, object detection captures objects' class bounding boxes to decide whether a position is based on abnormal activities. There are several popular object detector such as SSD [27], YOLOv3 [28] and Mask R-CNN [29]. SSD [27], YOLOv3 [28] have a faster processing speed but can only detect the objects' bounding boxes. In contrast, Mask R-CNN [29] can detect the objects' bounding boxes and contours at the same time. Considering the computational cost, we use the popular object detector SSD [27] to provide bounding boxes, that is, Region Of Interest (ROI), and labels for a wide range of objects. SSD is exploited to extract the critical object position from RGB data to be fed into the learning process for object position anomaly detection.

3 | PROPOSED JOINT RGB-POSE AND OBJECT POSITION MODEL FOR HAAD

This section proposes to exploit the models defined for HAAD. The proposed approach is *bi-level*, which means that it can detect two types of anomalies, that is, abnormal body movements and abnormal object positions.

3.1 | HAAD problem statement

Let $\mathbb{D} = \mathbb{N} \cup \mathbb{T}$ a video dataset, where $\mathbb{N}$ contains human action *normal* data while $\mathbb{T}$ represents *testing* data, which contains either normal or abnormal human actions. In particular, let $\mathbb{N} = \{\mathbf{s}_j\}_{j=1}^{n}$ be the normal video data subset containing n clips of arbitrary time length, where $\mathbf{s}_j$ represents the j-th RGB video clip. Let

$$\mathbb{T} = \left\{ \mathbf{v}_i, \{g_{\mathbf{v}_i}(t|\phi)\}_{\phi=1}^{d} \right\}_{i=1}^{m}, \tag{1}$$

be the video testing dataset containing m clips, where $\mathbf{v}_i$ represents the i-th RGB video clip, $g_{\mathbf{v}_i}(t|\phi)$ is the anomaly based ground truth given for the target identity index ϕ. In particular, $g_{\mathbf{v}_i}(t|\phi)$ is a function that associates each frame t and target identity ϕ to a binary response,that is, normal/abnormal label, and it can be defined as follows:

$$g_{\mathbf{v}_i}(t|\phi) = \begin{cases} 0, & \textit{if } \phi \textit{ performs normal action in } t \\ 1, & \textit{if } \phi \textit{ performs abnormal action in } t \end{cases} \tag{2}$$

Since $\mathbb{D}$ can be a single or a multi-target dataset, the target identity is provided by a tracking mechanism operating over the target detections.

Thus, the goal of this section is to propose a strategy to compute $G_{\mathbf{v}_i}(t|\phi) \approx g_{\mathbf{v}_i}(t|\phi)$ for all $i = 1, \ldots, m$ and ϕ, and simultaneously providing an explanation of the performed action, defined as $L_{\mathbf{v}_i}(t|\phi)$, in the form of an action label.

In principle, action anomalies within $\mathbb{T}$ might be due to the unexpected target's body movements or with an unexpected positioning/usage of contextual objects. Therefore, this work proposes an anomaly detection algorithm that can detect both anomalies (Figure 1).

3.2 | Body movements based HAAD

3.2.1 | Pose features extraction

The first step consists of extracting pose-related information from RGB data. To this purpose, OpenPose [14] is exploited to detect body landmarks of visible targets. This detector can provide up to 25 body landmarks and detection confidence for

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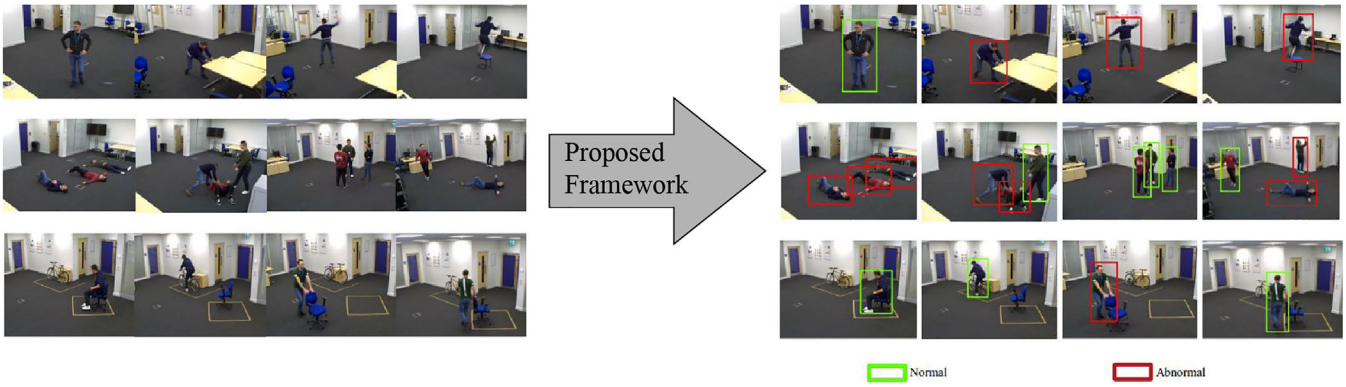

Methods \ Criteria	Anomaly Frame Labels	Detections in Frame Level	Bounding Boxes	Action Labels	Anomaly Labels for Bounding Boxes	Object Identities
Proposed Framework	✓	✓	✓	✓	✓	✓
Traditional Approaches for Anomaly Detection	✓	✗	✗	✗	✗	✗

FIGURE 1 Illustration of the studied problem and the motivation of the work

each landmark. The target bounding box can be retrieved by the detected landmarks set.

These trackers consider both target detections and target appearance provided by RGB data within the target's bounding box. In principle, since multiple targets can be visible in each frame, one of our current tracking systems [4, 15, 30] can be exploited for labelling each detection with a unique target identity index. However, for simplicity, a more straightforward tracking system only based on the target's ROI coordinates has been applied [31].

Therefore, the tracked landmark sequences are treated with the *ActionXPose*, *spatio-temporal-attention mechanism* [2]. This robust method can transform the landmark sequence into a meaningful multi-variate sequence that encodes spatial and temporal information. This method creates poses libraries for each training action and exploits them to encode meaningful features in the training phase.

Therefore, the output of this step consists of features in the form of multi-variate time sequences, encoding different aspects of the motion, including the mutual relation between sub-body parts and full-body similarities with the learned libraries.

3.2.2 | Proposed joint RGB-pose models

This section aims to propose three models for Joint RGB-Pose based action classification. The main idea is to jointly learn from both RGB data and pose-related features explained in Section 3.2.1, imposing a collaborative deep learning structure.

Theoretically, if I represents the aggregate information contained within RGB data, then the pose-related information P is such that $P \subset I$, because poses are in turn obtained from RGB data. Therefore, it is reasonable to argue that learning from RGB data is expected to give similar, if not superior, results to learning from poses. However, as shown in this work, this is not the case due to the deep learning limitations, for example, it does not allow to explicitly point the network's attention over the correct image regions according to the human perspective. We propose three joint approaches based on multimodal data and multimodal learning algorithms to highlight and overcome this limitation. The proposed approaches variety is needed to support our conclusions adequately.

Therefore, we defined three Joint RGB-Pose models, named Model A, B and C, deploying novel combinations of commonly used and well-established approaches for video-based deep learning, such as ConvNets, Two-streams networks and 3DCNNs [32].

As shown in Figure 3, Model A, B and C share a typical parallel-learning structure, where pose-based and RGB-based information flow in two deep learning branches. However, the proposed structure is not based on two completely independent branches. Generally, independent branches are those where the final decision is made by fusing the information at the score level [33]. However, this approach prevents from considering dependency between modalities. For example, it is reasonable to expect that pose-based learning is particularly effective for posture-related actions. At the same time, RGB data can describe finer appearance-related details, supporting the action recognition, especially when poses are not informative enough. However, challenging viewpoints can produce incomplete or corrupted poses. Therefore, it might be convenient to trust more RGB data than poses in these particular cases. In contrast, when poses are clear and body joints detectable, poses might be more effective as an information source. In other words, it is required to discriminate data sources dependently. Therefore, inspired by information fusion in [34], we incorporate this inter-dependency by firstly processing RGB data and poses separately. Subsequently, features vectors are concatenated, followed by a Fully Connected layer (FC) to learn dependency between feature vectors. Due to backpropagation, the network

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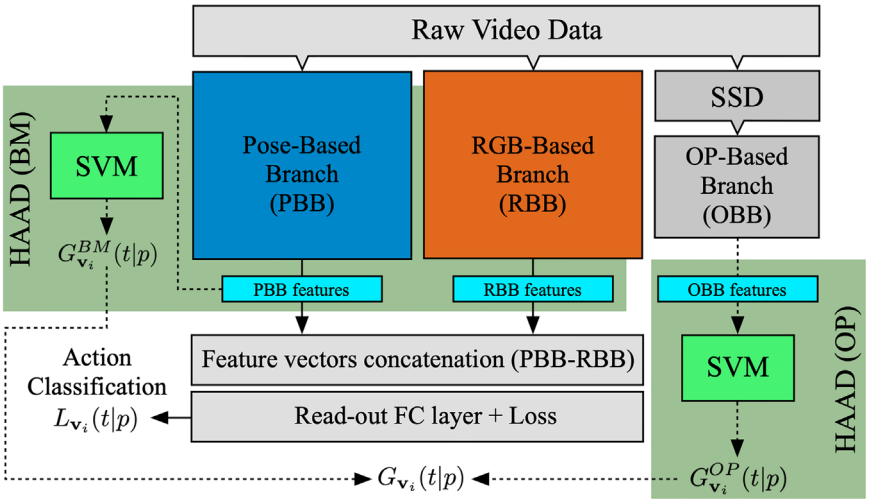

FIGURE 2 Proposed pipeline for bi-level HAAD. Colour version is recommended

can jointly optimize its weight according to the contribution of both modalities depending on the conditions. This learning structure is shown in Figure 3, where the three proposed methods are drawn.

Alongside the above-mentioned typical learning structure, it is necessary to specify how to extract feature vectors from each modality. Regarding the pose-branch, Models A, B and C are based on the Multivariate Long Short-Term Memory and Fully Convolutional Network (MLSTM-FCN) network [35], as supported by our extensive evaluation in [2]. Regarding the RGB-branches, inspired by [32], three different networks are considered:

- **CRNN** RGB data is processed through four CNN and two Fully Connected (FC) layers to extract frame-level features. Subsequently, the LSTM is used to model the time dependencies of the features across frames.
- **ResNetCRNN** RGB data is processed by using ResNet152 [36], pre-trained on ILSVRC-2012-CLS dataset [37], to extract frame-level features. Subsequently, LSTM is used to model the time dependencies of the features across frames.
- **3DCNN** RGB data is processed using two 3DCNN and 2 FC layers to extract frame-level features and model time dependencies.

Therefore, Model A, B and C are defined as in Figure 3, combining the RGB-branches mentioned above with the Pose-branch based on MLSTM-FCN. All models concatenate the features provided by the two parallel branches, followed by the usual FC layer to transform the combined feature vector into a class-based binary vector. Therefore, the loss computes the error between predicted classes and the ground truth.

A joint RGB-pose model (A, B or C) has been trained for Action classification on a pre-defined dataset (different than $\mathbb{D}$), with the unique restriction to consider k frames length action sequences. Let $\mathcal{M}_k$ be this model. Let also consider that the RGB data associated with the action sequences only consist of the target's bounding box. Therefore, we should assume that, to some extent, the model $\mathcal{M}_k$ can also perform action classification on unseen datasets. In particular, let consider $\mathbb{D}$ as the unseen dataset. Therefore, as shown in Figure 2, clips in $\mathbb{N}$ can be processed in the training phase to extract Pose-Based Branch features (PBB). Thus, PBB features are exploited to train an SVM one-class model for semi-supervised anomaly detection [38]. Similarly, $\mathbb{T}$ clips are processed in the testing phase to extract PBB features to be injected into the trained SVM model to determine whether the testing sequence should be considered normal or abnormal. Since each PBB feature vector corresponds to a certain time window $[t_1, \dots, t_k]$ for a given target ϕ, therefore $G^{BM}_{\mathbf{v}_i}([t_1, \dots, t_k]|\phi)$ is computed. The superscript BM denotes that G is obtained considering pose-based body movement-related features. In principle, RGB-Based features (RBB) can also be exploited for the same task. However, we prove in Section 4.2.4 that RBB features are not as effective as PBB features.

It is worth mentioning that, regardless of the human action labels included in the $\mathcal{M}_k$ training, $\mathcal{M}_k$ is always able to extract features from *any* and *unseen* k frames-length action sequence. Suppose the unseen sequence resembles one of the $\mathcal{M}_k$ training sequences. In that case, the resulting feature vector is such that the read-out FC layer can reasonably provide the expected action label $L_{\mathbf{v}_i}([t_1, \dots, t_k]|\phi)$. As opposite, if the unseen sequence does not resemble training data for $\mathcal{M}_k$, the resulting action label will be not necessarily an accurate description of the performed action. Nevertheless, the SVM model can still be trained on those features, regardless of the action-related knowledge they are carrying.

As the first necessary approximation, we assume that posture-related HAAD is primarily independent of the object position in this work. Therefore, in the next section, object position based HAAD is proposed as the additional and independent level of HAAD.

3.3 | Object position based HAAD

In this section, the goal is to perform anomaly detection based on the relative position between the target ϕ and the contextual Objects Position (OP). Let assume that the dataset $\mathbb{D}$ contain a given set of w key objects O (objects under monitoring,

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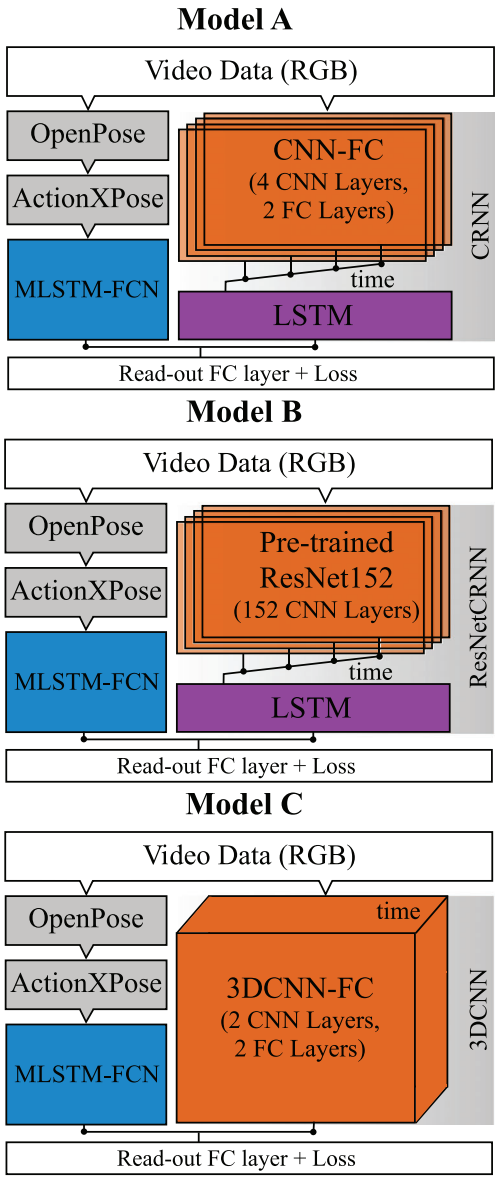

FIGURE 3 Joint RGB-Pose proposed models. In all models, Video RGB data is firstly pre-processed with OpenPose and ActionXPose to extract pose-related features to feed the pose-based learning branch performed by the MLSTM-FCN. In parallel, raw RGB data also feeds the RGB-branch, which is based on different structures depending on the model, that is, Model A, B and C. Feature vectors extracted by both branches are concatenated and subsequently transformed by an FC layer, before the loss computation. Colour version is recommended

including the human target). For each given frame t, let

$$\begin{aligned} &\textit{Human target}: \ (\phi_x, \phi_y) \\ &\textit{Other objects}: \ \left\{ (x,y)_{h,1}, \dots, (x,y)_{h,J_h} \right\}_{h=2}^{w} \end{aligned} \tag{3}$$

be the positions for the objects in O, where J_h is the number of occurrences of the h-th object in O. Therefore, anomalies might come from the unexpected mutual positions between different objects in O. Thus, for each human target ϕ and for each frame t, multiple position-based $\mathbf{q}(t,p) \subset \mathbb{R}^{2w}$ feature vectors can be defined, one for each occurrence, as follows:

$$\begin{aligned} &\mathbf{q}(t,p) = \left\{ [\phi_x, \phi_y, x_{2,j_2}, y_{2,j_2}, \dots, x_{w,j_w}, y_{w,j_w}] \right\} \\ &\forall j_2 \in [1, \dots, J_1], \dots, \forall j_w \in [1, \dots, J_w] \end{aligned} \tag{4}$$

The advantage of this approach is that it allows straightforward generalization to multiple human targets and multiple occurrences of the same object type in the scene. Let the Object Position-based Branch (OBB) be the set of OP feature vectors, as in Figure 2, that is, $\mathbf{q}(t,p) \subset$ OBB for a given frame t and target ϕ. Similarly to the previous section, a one-class SVM model can be trained by using normal features OBB extracted from $\mathbb{N}$ and tested OBB features extracted from $\mathbb{T}$. However, as opposed to the previous section where k frames were considered at a time, the abnormality estimation is based on *single* frame by using multiple feature vectors as follows:

$$G_{\mathbf{v}_i}^{OP}(t|\phi) = \begin{cases} 0 & \textit{if } SVM(\mathbf{q}(t,\phi)) = 0 \ \ \forall \mathbf{q}(t,p) \\ 1 & \textit{if } \exists \mathbf{q} \in \mathcal{Q}(t,\phi) \ s.t. \ \ SVM(\mathbf{q}(t,\phi)) = 1 \end{cases} \tag{5}$$

where SVM($\mathbf{q}(t,\phi)$) denotes the trained SVM model tested on the vectors $\mathbf{q}(t,\phi)$. Figure 2 depicts the pipeline of this step as part of the more general bi-level HAAD.

3.4 | Bi-level BM-OP activity anomaly detection

Based on the assumption made in Section 3.2 regarding BM and OP independency, in this section we only consider that BM and OP based anomaly detections, that is, $G_{\mathbf{v}_i}^{BM}(t|\phi)$ and $G_{\mathbf{v}_i}^{OP}(t|\phi)$, can be simply combined by using a simple *Logic OR*, as follows:

$$G_{\mathbf{v}_i}(t|\phi) = \begin{cases} 0 & \textit{if } G_{\mathbf{v}_i}^{BM}(t|\phi) = 0 \wedge G_{\mathbf{v}_i}^{OP}(t|\phi) = 0 \\ 1 & \textit{otherwise} \end{cases}, \tag{6}$$

where $G_{\mathbf{v}_i}(t|\phi)$ is the aimed estimation of the anomaly detection ground truth for the $\mathbf{v}_i$ video clip. See Figure 2 for a general overview of the proposed bi-level approach.

To the best of our knowledge, there are no standard methods to provide ROC curves for SVM one-class based anomaly detection models. However, the ROC curve is a standard metric for anomaly detection systems and allows effective comparisons between different models performance. A simple solution to obtain ROC curves for SVM one-class models is proposed. A smooth (not binary) decision surface is required to compute ROC curves. However, SVM one-class model does not output a smooth decision surface since it is impossible to fit a logistic function by using only one-class data properly. Therefore, we propose to weaken this requirement by imposing the logistic function to be part of the anomaly detection model.

Let $s \in [-\infty, \infty]$ be the obtained SVM score for testing data. We propose to simply transform s by using a logistic function as

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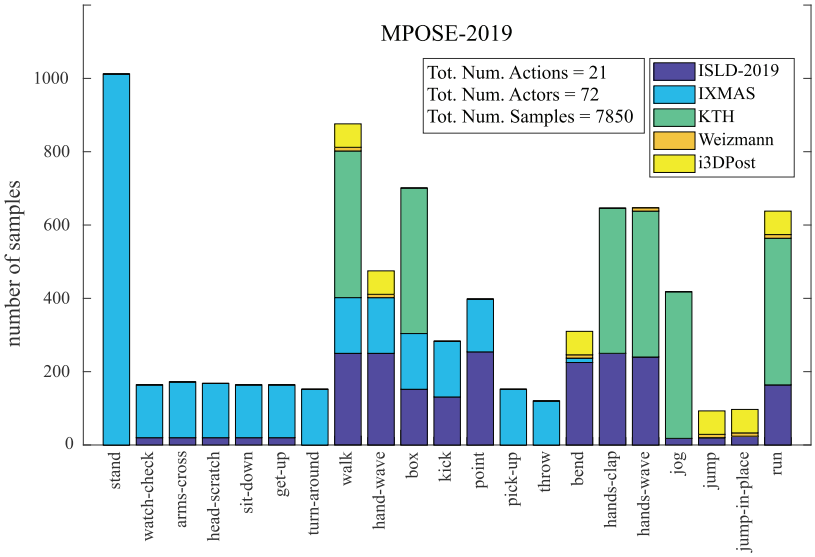

FIGURE 4 MPOSE-2019 actions statistics. MPOSE-2019 contains 7850 video clips collected from famous posture-based action classification datasets. Video clips contains 21 posture-related actions performed by 72 human targets. Colour version is recommended

follows:

$$f(s) = \frac{1}{1 + e^{-k_0(s-s_0)}}, \tag{7}$$

where s_0 is the sigmoid midpoint and k_0 is the curve steepness. Samples such that $s = 0$ lay on the edge of the SVM decision boundary. Conversely, samples such that $s > 0$ or $s < 0$ lay, respectively, inside or outside the normal region. Therefore, it is natural to set $s_0 = 0$, in order to obtain $f(s = 0) = 0.5$, $f(s > 0) > 0.5$ and $f(s < 0) < 0.5$. The parameter k is inversely proportional to the confidence of the model and can be a-priori fixed. The defined $f(s)$ represents the new score function which allows the ROC curve computation. This definition of $f(s)$ does not include any learnable parameter and can be merged as part of the SVM model when ROC curves are required.

4 | SIMULATIONS AND RESULTS

4.1 | Datasets

4.1.1 | MPOSE-2019 dataset

In this section, Model A, B and C are tested on a challenging dataset called MPOSE-2019. MPOSE dataset has been presented the first time in our recent publication [2]. In summary, MPOSE contains video data collected from five different datasets, that is, KTH [39], i3DPost [40], IXMAS [41], Weizmann [42] and ISLD [2], devided in 21 action classes. In this work, we further extend the MPOSE, defining MPOSE-2019. This extension include additional sequences for some actions, that is, *walking*, *hand-waving*, *boxing*, *pointing*, *bending*, *hands-clapping*, *hands-waving* and *running*, to extend the viewpoints variability. In Figure 4, a summary of the action composition of MPOSE-2019 is shown.

As discussed in [2], MPOSE-2019 represents the best scenario where ActionXPose can be effectively used, making the most from it since the target body is potentially fully visible. Therefore, in this section, ActionXPose is effectively used to transform pose data provided by OpenPose into meaningful temporal-motion features, as discussed in Section 3.2.1. For strategies for dealing with body occlusions, [2] provides robust solutions which are beyond the purpose of this work.

4.1.2 | BMbD, M-BMbD and JBMOPbD datasets

In literature, there is a multitude of datasets for video-based anomaly detection. UCF-Crime [6] is a challenging dataset that includes multiple abnormal instances from several and different semantic events. However, the proposed human-related abnormal classes are limited and mostly not related to posture. Therefore, this dataset is not suitable as a benchmark for a pose-based method. Other datasets such as Hockey Fight, Subway, UMN, Violence in Crowds (VIC), Violence in Movies (VIM) and BEHAVE [43–46] are also not suitable for highlighting the efficacy of the proposed pose-driven HAAD, that is, classifying posture-related actions and detecting related abnormal events. The first limitation regards the lack of explicit human posture-based anomalies. Targets bodies are mostly out of the camera field of view or, even if fully visible, the targets are not performing abnormal actions in terms of postures. The second limitation is that, in most cases, the available RGB resolution is very low, and the human pose detection is compromised to allow the proposed pose-driven HAAD, especially in low quality or highly cluttered background frames. The third limitation is related to objects. Indeed, visible objects positions do not explicitly define an anomaly detection problem that can show the efficacy of the proposed method and the limitations of state-of-the-art approaches. For these reasons, in this work, novel challenging datasets, BMbD, M-BMbD and JBMOPbD were recorded, focusing on posture-related and object position-related anomalies.

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FIGURE 5 BMbD, M-BMbD and JBMOPbD datasets examples. Green bounding boxes depict the normal events, while the red bounding boxes represent abnormal events. Colour version is recommended

The recording set up includes two standard RGB cameras mounted in a fixed position on two top-corner of the Intelligent Sensing Lab, Newcastle University, UK[2]. The dataset includes spontaneous actions performed by up to 3 actors, wearing different clothes to include appearance change challenges. Additionally, three types of objects are considered, that is, human target, a chair, and a bike, for OP-based anomaly detection. The human targets are also free to interact with other environmental objects, such as drawers, tables, monitors, and other human targets. Video data within the dataset is divided into *normal* and *testing*, to allow semi-supervised anomaly detection, as mentioned in the problem statement in Section 3.1. The performed action anomalies consist of (1) unexpected body movements, not necessarily among a pre-defined set of actions; (2) key object position anomalies. In particular, regarding JBMOPbD normal dataset, the position of the key objects was constrained to be within rectangular marks on the floor. In the JBMOPbD testing subset, the targets were free to move the objects in any position (Figure 5).

Table 1 summarizes the abnormal semantic domains for each proposed dataset to independently and jointly evaluate BM, OP, single and multi-target performances.

The ground truth for testing videos in ISLD-A has been manually set for HAAD. Therefore, for each tracked human target, a binary label normal/abnormal has been fixed for each target bounding box, frame-by-frame. Regarding action classification, no ground truth has been set since the performed actions were spontaneous and challenging to be always clearly classified.

[2] http://www.intellsensing.com/research/information-processing/multimodal-surveillance/

As evaluation metric for HAAD, we suggest to consider the overall frame-level average accuracy, which measures the following ratio

$$\frac{1}{N}\sum_{t=1}^{N}\frac{TP(t)+TN(t)}{P(t)+N(t)}, \tag{8}$$

where $TP(t)$ and $TN(t)$ represent, respectively, the true positive (abnormal) and true negative (normal) detections at frame t, $P(t)$ and $N(t)$ represent the real positive (abnormal) and real negative (normal) detections at frame t, for all available frames N. When multiple human targets are available on the same frame t, thus $TP(t)$, $TN(t)$, $P(t)$ and $T(t)$ are cumulative over visible targets. As additional metric, ROC curves are considered to allow effective state-of-the-art comparisons.

4.2 | Bi-level evaluations for HAAD

4.2.1 | Body movement performance

As mentioned in Section 3, the combined HAAD model trains its action classification abilities on a different dataset. MPOSE-2019 is used for this purpose since it best fits the criteria for HAAD on the novel datasets introduced in the previous section. It is worth underlining that actions into MPOSE-2019 are mainly posture-related and unrelated to the background. For this work's purpose, background information has been further decreased by considering only the target's ROIs, neglecting the rest of the frame background. The Joint RGB-Pose

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TABLE 1 ISLD-A splits features summary

Dataset	Frames					Viewpoints	Maximum Targets	Abnormal Events
	Total	Training	Testing	Regularity	Irregularity			
BMbD	44220	23432	20788	34130	10090	2	1	58
M-BMbD	40918	24088	16830	36648	4270	2	3	43
JBMOPbD	38240	24010	14230	31038	7202	2	1	32

TABLE 2 MPOSE-2019 dataset action classification performance. Baseline models, that is, the pose-based MLSTM-FCN and the RGB-based models, are compared with the proposed Joint RGB-Pose models, that is, Model A, B and C

	Modalities	Acc	Acc
MLSTM-FCN	Poses	90.3%	n/c
CRNN	RGB	90.4%	n/c
Model A	RGB-Poses	n/c	94.9%
ResNetCRNN	RGB	80.8%	n/c
Model B	RGB-Poses	n/c	**95.8%**
3DCNN	RGB	77.3%	n/c
Model C	RGB-Poses	n/c	90.4%

models outperform the pose-based and RGB-based models. This result further confirms the validity of the multimodal approach in action classification. Moreover, as we have seen in Section 4.2.1, Model B achieves the best action classification results on MPOSE-2019. Therefore, Model B is chosen as the central architecture to test the bi-level HAAD strategy (Table 2).

4.2.2 | HAAD Performance

This study provides details on how to train the HAAD model. Let Model B be the chosen architecture, denoted as $\mathcal{M}_k$, and trained on MPOSE-2019. The knowledge learned from MPOSE-2019 for action classification is exploited to perform HAAD on other datasets, for example, BMbD, M-BMbD and JBMOPbD. As discussed in Section 3.2, we impose MPOSE-2019 training/validation data do be randomly cropped by using an experimentally set 30-frames time window, that is, $k = 30$. Therefore, training samples are randomly cropped at each training epoch to increase generalization. As opposed, validation samples are randomly cropped only at the beginning of the training to keep stable validation data over different epochs and consistently improve the validation curve. Figure 6a shows the validation curve obtained during the pose-branch human actions training, where only epochs $\mathcal{E} \subset \{1, \dots, 100\}$ where the validation score increased are reported on the x-axis. The status of the network has been saved at each epoch $e \in \mathcal{E}$, obtaining models $\mathcal{M}_{30}^{e}$, for all $e \in \mathcal{E}$. Therefore, for all $\mathcal{M}_{30}^{e}$, the HAAD corresponding accuracy on the HAAD datasets are reported on Figure 6-b)-e), for all dataset splits. Since BM related HAAD is based on a SVM one-class model which requires to set an additional parameter α, that is, the *outlier fraction*, we reported the accuracy obtained for $\alpha = \{.01, .04, .07, .10, .13, .16, .19, .22\}$. Regarding the outlier fraction parameter β required for OP based HAAD, it has been experimentally set to $\beta = .01$.

For BMbD and M-BMbD, the peak is at, respectively, the 23rd and 24th epochs. For the JBMPObD, the peak is on the 17th epoch when OP-based HAAD is not considered and on the 18th when OP-based HAAD is considered. This surprising behaviour might be due to the concurrence of two major causes. First, the BM features are not designed to optimize the SVM performance but the action-based loss function. While the action classification performance follows an increasing monotonic curve, the HAAD performance might be not necessarily monotonic. Second, there might be a slightly overfitting effect due to the chosen training strategy, which randomly crops training sequences. In fact, as the action classification validation score on MPOSE-2019 increases, the sequence random cropping might force completely different sequences to have the same output in terms of features. For example, data that resemble a *standing* action often surrounds other actions in the same sequence. Therefore, randomly cropping the sequence around it forces data resembling the *standing* action to be labelled differently. This strategy improves generalization for action classification but might reduce the HAAD features sensitivity.

Regarding α, as expected, the parameter plays an essential role in optimizing performance and needs to be chosen according to the tested dataset split for best performance.

4.2.3 | State-of-the-art comparisons

In this section, comparisons between Model B, ResNetCRNN and official implementation of available algorithms [8, 9] for video-based anomaly detection is provided. The chosen benchmarks are BMbD and JBMOPbD splits. [8, 9] provide ROC curves as standard performance evaluation metric. Therefore, we implemented the ROC curve for SVM one-class models as proposed in Section 3.4. Regarding Model B and ResNetCRNN, ROC curves for (Figure 7)

$$\alpha = [.01, .04, .07, .1, .13, .16, .19, .22], \tag{9}$$

and

$$k_0 = [.1, .2, .3, .4, .5, .6, .7, .8], \tag{10}$$

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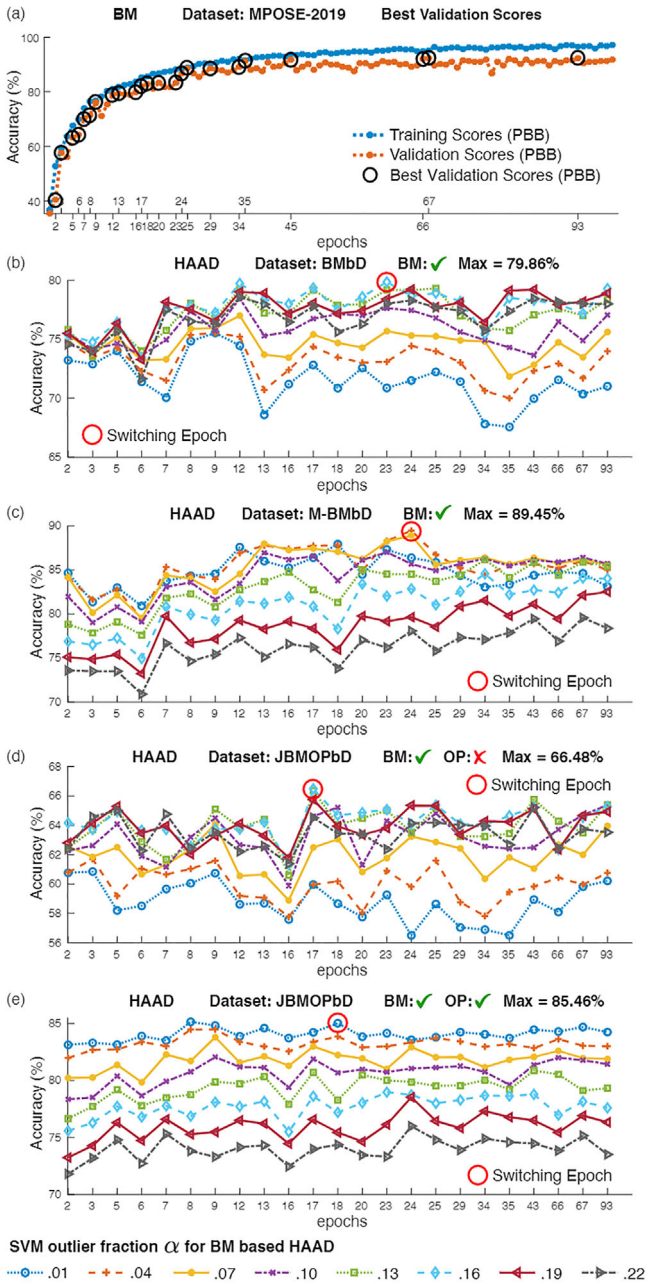

FIGURE 6 Cross-datasets (MPOSE-2019 to ISLD-A) HAAD epoch-based study for switching epoch (○) research. (a) Model B pose-branch best validation scores. (b) Corresponding HAAD accuracy for BMbD. (c) Corresponding HAAD accuracy for M-BMbD. (d) Corresponding HAAD accuracy for JBMOPbD, in the case of disabled OP based HAAD. (e) Corresponding HAAD accuracy for JBMOPbD, in the case of enabled OP based HAAD

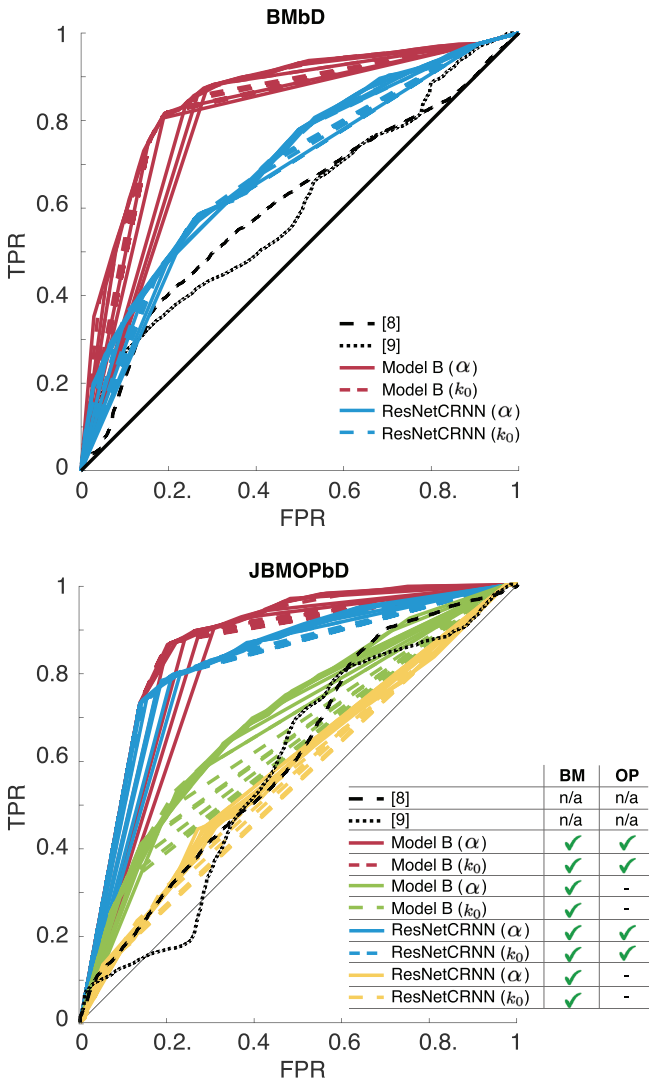

FIGURE 7 Model B, ResNetCRNN and state-of-the-art methods performance comparison. (Top) ROC curves for the BMbD dataset obtained by varying parameter $\alpha = [.01, .04, .07, .1, .13, .16, .19, .22]$ and $k_0 = [.1, .2, .3, .4, .5, .6, .7, .8]$. (Bottom) ROC curves for the JBMOPbD dataset obtained by varying parameter $\alpha = [.01, .04, .07, .1, .13, .16, .19, .22]$ and $k_0 = [.1, .2, .3, .4, .5, .6, .7, .8]$, and enabling/disabling BM/OP anomaly detection. Colour version is recommended

are drawn. In the case of JBMOPbD, BM and OP based HAAD are selectively enabled/disabled, to highlight the importance of both levels of analysis.

Obtained results show that the proposed Model B significantly improve performance over the state-of-the-art autoencoders and ResNetCRNN baselines. These results confirm that autoencoders-based HAAD fails on the challenging problems mentioned in this work because the abnormality is not merely related to significant changes in the RGB content. When the human target starts performing an abnormal action, the target appearance remains mostly preserved, preventing the autoencoder from failing in reconstructing the RGB data. In contrast, the proposed approach relies on contextual and pre-trained knowledge, effectively transferred to the HAAD problem.

4.2.4 | HAAD ablation study

In this section, the ablation study for HAAD is provided. This study highlights the contribution of different features to the overall HAAD accuracy. In particular, all relevant combinations of features are systematically exploited following the methodology presented in Section 4.2.2. Therefore, for each dataset,

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combinations of pose-based branch features, that is, PBB, and RGB-based branch features, that is, RBB, are considered for the BM based HAAD. Moreover, for the JBMOPbD dataset, BM based features are considered alongside OP based features, that is, OBB, in all possible setups. The best accuracy over epochs $\mathcal{E}$ is reported in Table 3 for each case, including relative α and β parameters. As a useful comparison, in this study, the ResNetCRNN performance as HAAD features extraction method is also reported. To this purpose, ResNetCRNN has been trained on MPOSE-2019 using the same 30-frames random time cropping already used for the Model B training. The obtained training process is depicted in Figure 8. Following the same approach already used for Model B, several network statuses are saved using the best validation scores. Therefore, resulting networks are used for HAAD features extraction.

The overall conclusion of this study is that RGB related features (RBB) are constantly worsening the HAAD results and should not be considered, supporting the study reported in Section 4.2.2. On the other hand, considering OP related features (OBB) is always beneficial, as expected, since the JBMOPbD dataset includes significant anomalies from the unexpected key object positions.

TABLE 3 HAAD Performance Ablation Study. Model B best accuracy and α parameter are reported for relevant combinations features, that is, PBB, RBB and OBB for HAAD. Results are compared with those obtained by the model ResNetCRNN, which is the precursor of the Model B

BMbD	**Features**				
model	**BM**	**Best accuracy**	α		
Model B	PBB	**79.86%**	.16		
	RBB	57.84%	.13		
	PBB+RBB	78.06%	.16		
ResNetCRNN	RBB	66.68%	.16		
M-BMbD	**Features**				
model	**BM**	**Best accuracy**	α		
Model B	PBB	**89.45%**	.04		
	RBB	74.10%	.04		
	PBB+RBB	88.50	.07		
ResNetCRNN	RBB	86.17%	.07		
JBMOPbD	**Features**				
model	**BM**	**OP**	**Best accuracy**	α	β
Model B	n/c	OBB	81.80%	n/c	.01
	PBB	n/c	66.48%	.16	n/c
	PBB	OBB	**85.46%**	.01	.01
	RBB	n/c	56.41%	.19	n/c
	RBB	OBB	81.46%	.01	.01
	PBB+RBB	n/c	65.20%	.13	n/c
	PBB+RBB	OBB	85.11%	.04	.01
ResNetCRNN	RBB	n/c	57.31%	.13	n/c
	RBB	OBB	82.76%	.01	.01

4.3 | Critical analysis and discussion

The goals of these extensive studies are to highlight that: (1) pose-driven action classification (Joint RGB-Pose learning) bring benefits over the single modality based learning; (2) state-of-the-art autoencoders-based HAAD fails in challenging problems such as those considered in this work; (3) the proposed pose-driven action classification structure can be used for HAAD, exploiting its ability to extract meaningful feature out of actions samples, outperforming its RGB-based feature extraction counterpart.

This approach creates a novel link between abnormal behaviours and positions for HAAD. However, further studies are required. For example, it has been proven that the best HAAD performance does not correspond to the best validation epoch during action classification training. This is counterintuitive and troublesome, forcing the action classification model and HAAD validations to be performed simultaneously during the training. This undesirable feature can, in principle, be surpassed by investigating which features characteristics allow SVM performance to be optimal. Therefore, by embedding additional layers within the action classification structure to replicate such characteristics through network convergence, we would be reasonably sure that HAAD and the inside action classification accuracy would both follow the same monotonic growing trend.

However, the primary purpose of this work was to trace a direction. More action classes would be desirable for practical implementation in a real-world scenario and more extensive posture-based HAAD datasets (e.g., including more objects and targets in even more challenging situations). Recording new action and additional anomaly detection data are expensive and highly time-consuming, especially for annotating clips frame-by-frame. MPOSE-2019 has been created to mitigate these issues, taking the most from famous and already available data.

Ultimately, despite the limitations mentioned above, the effectiveness of the proposed approach over the baselines in HAAD is the most significant contribution. In particular, it is crucial to highlight that the novel joint HAAD proposed approach allows HAAD sequences to be *captioned* with action classification labels in a multi-target real-world setting, which means the proposed framework can also be used in other video-based vision problem, such as action localization [47]. Possible applications on healthcare, surveillance and automated alarm are straightforward.

5 | CONCLUSIONS

This work demonstrated that state-of-the-art, fully RGB-based approaches for semi-supervised anomaly detection have limitations on detecting human posture-related anomalies. They fail in detecting objects positions related anomalies, as well as abnormal body postures. Moreover, we demonstrated that pose-driven networks has the potential to outperform the RGB-based approaches in detecting abnormal postures, and boost performance in case of action recognition. Furthermore, a bi-level

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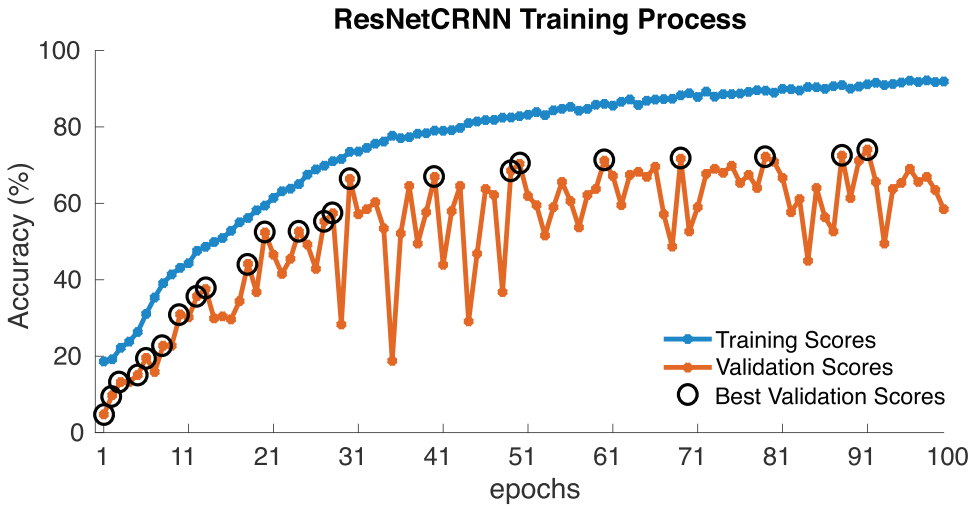

FIGURE 8 Training process for ResNetCRNN based on MPOSE-2019 using the 30-frames random time windows. The best validation network status is saved for HAAD features extraction. Best results obtained over those status are reported in Table 3. Colour version is recommended

HAAD approach has been proposed to simultaneously output: (1) body-movements anomaly detection labels; (2) object position anomaly detection labels; (3) action classification labels for single and multi-target video data. Future work will first concentrate on testing the proposed approach on various additional contexts for healthcare and surveillance applications. Second, this work raises further attention over the current trend in deep learning research, that is, multi-tasking network training strategies and fusion mechanisms. Therefore, future work will explore a novel fusion mechanism for multimodal and multi-tasking learning.

AUTHOR CONTRIBUTIONS

Yuxing Yang: Methodology, resources, software, validation, visualization, writing - original draft. Federico Angelini: Investigation, methodology, resources, validation, writing - original draft. Syed Mohsen Naqvi: Data curation, formal analysis, funding acquisition, project administration, supervision, writing - review and editing.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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